$C_{u\phi_2}^{(21)}$ i1_i2_ $\rightarrow$

$$\frac{1}{16\pi^2}\Big(\frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 y_u^{i1i2} LF_{3,2,0}[\tilde{\mu}, m_1] - \frac{1}{2} g_1^4 m_1^2 \tilde{\mu}^2 y_u^{i1i2} LF_{4,2,-1}[\tilde{\mu}, m_1] + \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 y_u^{i1i2}$$

$$LF_{3,2,0}[\tilde{\mu}, m_2] - 2 g_2^4 m_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{4,2,-1}[\tilde{\mu}, m_2] + \frac{1}{6} g_1^4 m_1^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,2,1,0}\left[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}\right] -$$

$$\frac{2}{3} g_1^4 m_1^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,2,1,0}\left[m_1, \tilde{\mu}, m_{\tilde{u}}^{i2}\right] + 2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,2,1,0}[m_2, \tilde{\mu}, m_1] -$$

$$2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{3,2,1,-1}[m_2, \tilde{\mu}, m_1] - \frac{3}{2} g_2^4 m_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,2,1,0}\left[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}\right] +$$

$$m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{3,1,1,0}[\tilde{\mu}, m_1, m_2] - \frac{3}{2} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{4,1,1,-1}[\tilde{\mu}, m_1, m_2] -$$

$$2 m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] + \frac{2}{9} g_1^4 m_1^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{3}{4} \tilde{\mu}^2 y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] -$$

$$\frac{3}{4} \tilde{\mu}^2 y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{1}{4} \tilde{\mu}^2 y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}\right] -$$

$$\frac{1}{4} \tilde{\mu}^2 y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}\right] -$$

$$\frac{1}{2} \tilde{\mu}^2 y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} LF_{2,1,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}\right] +$$

$$\frac{1}{2} \tilde{\mu}^2 y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} LF_{3,1,1,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}\right] -$$

$$\frac{3}{2} \tilde{\mu}^2 y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} LF_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{2} \tilde{\mu}^2 y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] +$$

$$\frac{3}{2} \tilde{\mu}^2 y_d^{sr} y_u^{pt} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_u}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}\right] -$$

$$\frac{3}{2} \tilde{\mu}^2 y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \overline{a_u}^{st} LF_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{2} \tilde{\mu}^2 y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \overline{a_u}^{st} LF_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] +$$

$$\frac{3}{2} \tilde{\mu}^2 y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \overline{a_u}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{3}{4} \tilde{\mu}^2 y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \overline{a_u}^{st} LF_{2,2,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] -$$

$$\frac{3}{2} \tilde{\mu}^2 y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \overline{a_u}^{st} LF_{2,1,1,1,0}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] +$$

$$\frac{3}{2} \tilde{\mu}^2 y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \overline{a_u}^{st} LF_{3,1,1,1,-1}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] -$$

$$\frac{1}{3} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,1,1,1,0}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i1}\right] -$$

$$\frac{2}{3} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{2,1,1,1,0}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{u}}^{i2}\right] +$$

$$\frac{1}{2} m_1 g_1^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] +$$

$$m_2 g_2^2 \tilde{\mu}^2 y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} LF_{2,1,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] +$$

$$\frac{1}{2} m_2 g_2^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{2,1,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] -$$

$$\frac{2}{3} m_1 m_2 g_1^2 g_2^2 \tilde{\mu}^2 y_u^{i1i2} LF_{1,1,1,1,1,0}\left[m_1, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] +$$

$$\frac{2}{3} m_1 g_1^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] -$$

$$\frac{1}{6} m_1 g_1^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] -$$

$$\frac{2}{9} m_1 g_1^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] +$$

$$m_2 g_2^2 \tilde{\mu}^2 y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} LF_{1,1,1,1,1,0}\left[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}\right] +$$

$$\frac{1}{2} m_2 g_2^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}\right] -$$

$$\frac{8}{3} m_3 g_3^2 \tilde{\mu}^2 y_u^{pi2} y_u^{i1r} \overline{a_u}^{pr} LF_{1,1,1,1,1,0}\left[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}\right] -$$

$$\tilde{\mu}^2 y_d^{pt} y_d^{i1r} y_u^{si2} \overline{a_d}^{pr} \overline{a_d}^{st} LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, \tilde{\mu}\right]\Big)$$